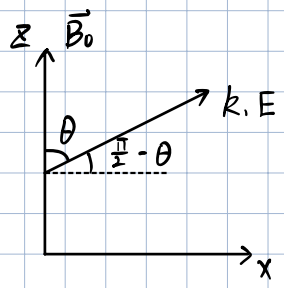
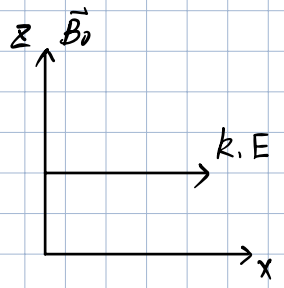
垂直 $\vec{B}$ 的静电离子波.





· $\vec{k}$ 几乎与 X 垂直.

$$\chi = \frac{\pi}{2} - \theta \propto \frac{v_{11,i}}{v_{11,e}} \propto \sqrt{\frac{m}{M}}$$

离子质量很大，不能实现德拜屏散.

$$\begin{cases} \vec{v}_0 = \vec{E}_0 = 0 \quad \vec{B}_z = \vec{B}_0 \\ T_i = 0 \\ \text{取 } \vec{E} = E_1 \hat{x} \quad E = -\nabla \phi_1 \end{cases}$$

$\chi = \frac{\pi}{2} - \theta$ $\quad M\vec{a} = e\vec{E} + e\vec{v} \times \vec{B}_0$

$$M\frac{\partial \vec{v}_{i1}}{\partial t} = -e\nabla\phi_1 + e\vec{v}_{i1} \times \vec{B}_0$$

$$\Rightarrow \begin{cases} -i\omega M \vec{v}_{ix} = -eik\phi_1 + ev_{iy}B_0 \\ -i\omega M \vec{v}_{iy} = -ev_{ix}B_0 \end{cases}$$

$$v_{iy} = \frac{eB_0 v_{ix}}{i\omega M}, \quad -i\omega M v_{ix} = -eik\phi_1 + \frac{e^2 B_0^2 v_{ix}}{i\omega M}$$

联立解得：$v_{ix} = \frac{-ekwM\phi_1}{e^2B_0^2 - \omega^2M^2} = \frac{ek\phi_1}{\omega M}\frac{1}{1-(\frac{eB_0}{\omega M})^2}$

$= \frac{ek\phi_1}{\omega M}\frac{1}{1-(\Omega_c/\omega)^2}$

考虑连续性方程：

电子：$\frac{\partial n_e}{\partial t} + \nabla\cdot(n_e\vec{v}_e) = \frac{\partial n_1}{\partial t} + n_0\nabla\cdot\vec{v}_1 = 0$

$\Rightarrow \quad -i\omega n_1 = -n_{0i}kv_1$

离子：$n_{i1} = n_0\frac{k}{\omega}v_{ix}$

$\frac{n_{e1}}{n_0} = \frac{e\phi_1}{kT_e} \Rightarrow \phi_1 = \frac{n_{e1}}{n_0}\frac{kT_e}{e}$

近似 $n_i = n_e$，$\frac{kT_e}{en_0}\frac{n_0k}{\omega}v_{ix} = \phi_1$

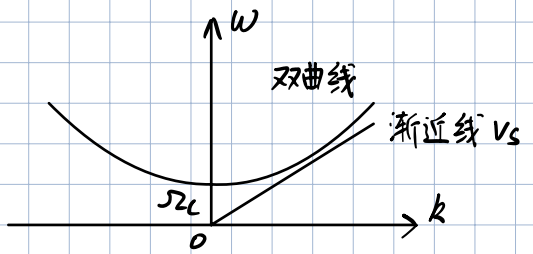
代回 $\vec{v}_{ix}$，有：

$(1-\Omega_c^2/\omega^2)\vec{v}_{ix} = \frac{ek}{\omega M}\frac{KT_e}{en_0}\frac{n_0k}{\omega}v_{ix} = k^2\frac{KT_e}{\omega^2M}$

$\omega^2 - \Omega_c^2 = k^2\frac{KT_e}{M} \Rightarrow \omega^2 = \Omega_c^2 + k^2v_s^2 \quad (v_s = \sqrt{\frac{KT_e}{m}})$

$v_p = \frac{\omega}{k} = \sqrt{(\frac{\Omega_c}{k})^2 + v_s^2}$

$v_g = \frac{d\omega}{dk} = \frac{v_s^2}{\sqrt{(\frac{\Omega_c}{k})^2 + v_s^2}}$



when $\theta \sim \frac{\pi}{2}$. Boltzmen 分布 X. 全运动方程.

$$mn\left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right] = qn(\vec{E} + \vec{v}\times\vec{B})$$

$$v_{ix} = \frac{ek}{\omega M}\phi_1 \frac{1}{1-(\Omega_i/\omega)^2}$$

$$v_{ex} = \frac{-ek}{m\omega}\frac{1}{1-(\omega/\omega_c)^2}$$

$$\begin{cases} n_{i1} = n_0 \frac{k}{\omega} v_{i1} \\ n_{e1} = n_0 \frac{k}{\omega} v_{e1} \end{cases}$$

$n_{i1} = n_{e1}$, $v_{i1} = v_{e1}$

$$M\frac{1}{1-\Omega_i^2/\omega^2} = -m\frac{1}{1-\omega_c^2/\omega^2} \Rightarrow \omega^2(M+m) = m\omega_c^2 + M\Omega_c^2$$

$\omega_c = \frac{eB}{m}$, $\Omega_c = \frac{eB}{M}$,

$$e^2B^2\left(\frac{1}{m}+\frac{1}{M}\right) \Rightarrow \omega^2(M+m) = \frac{e^2B^2(M+m)}{Mm}$$

$\therefore$ $\omega^2 = \frac{e^2B^2}{mM} = \omega_c\Omega_c$ 下杂化频率.

$$\begin{cases} v_p = \frac{\omega}{k} = \frac{\omega_c\Omega_c}{k} \\ v_g = 0 \end{cases}$$